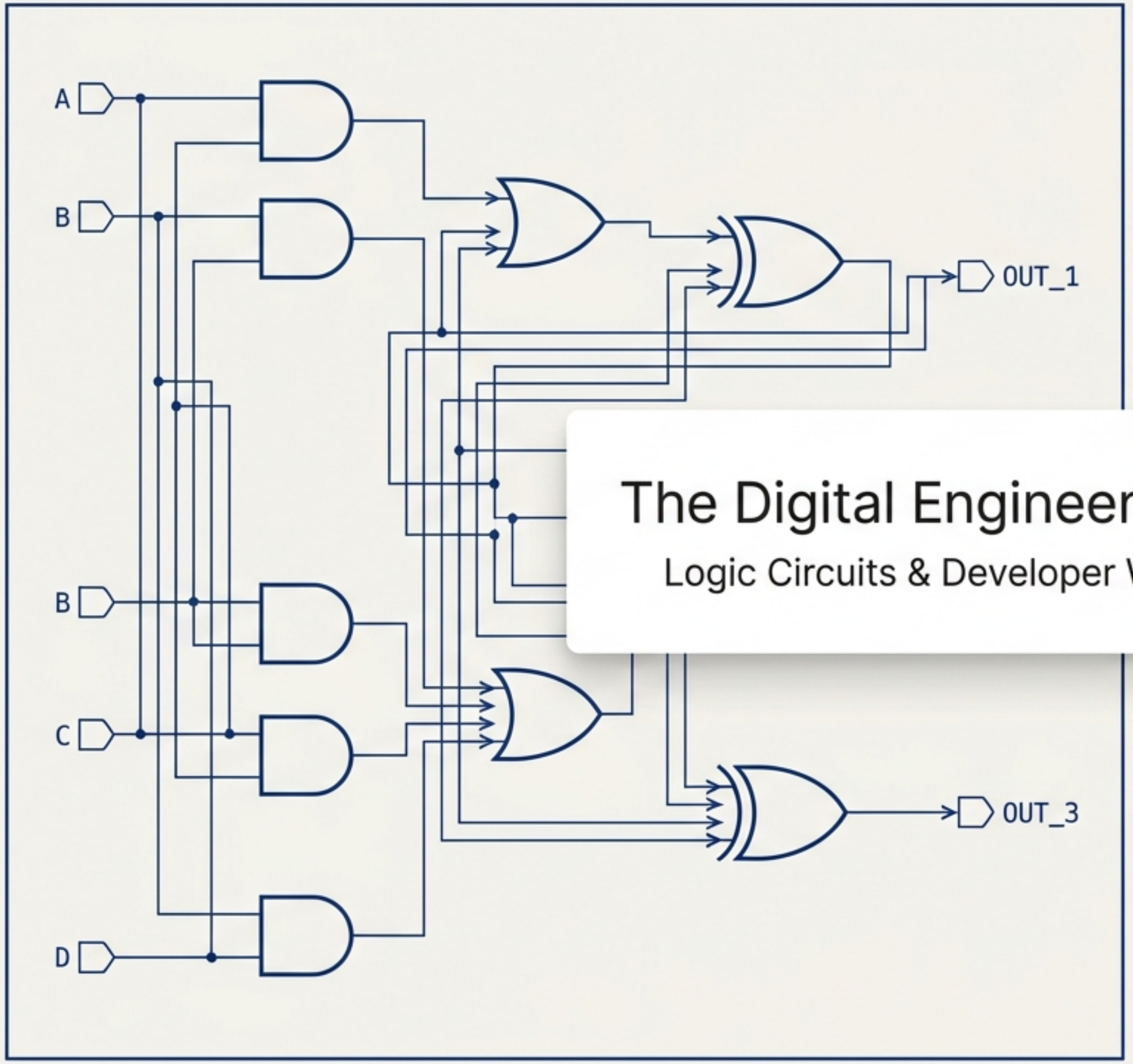


The Digital Engineer's Toolkit

Logic Circuits & Developer Workflows

logic_config.yaml — src/circuits/

logic_config.yaml ×

```yaml
# Digital Logic Configuration
# Project: Integrated_Circuit_v1.2

environment:
  mode: simulation
  target: FPGA_Xilinx_Artix-7
  clock_speed: 100MHz

modules:
  - name: AND_Gate_Block
    inputs: [A, B]
    1
    ay_ns: 0.5, power_mW: 1.2 }

    e_Block
    ]
    2
    ay_ns: 0.7, power_mW: 1.5 }

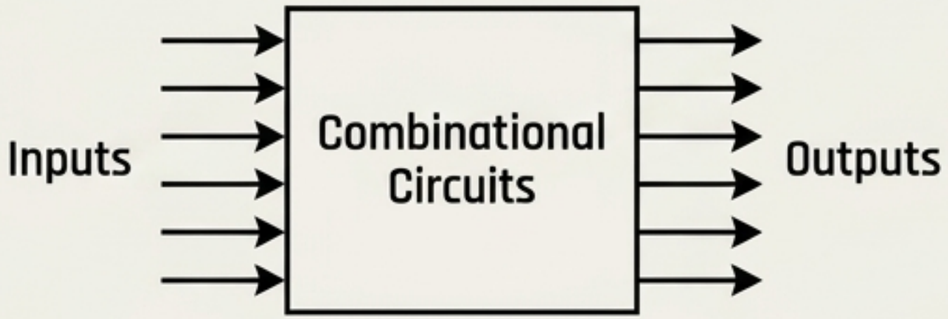
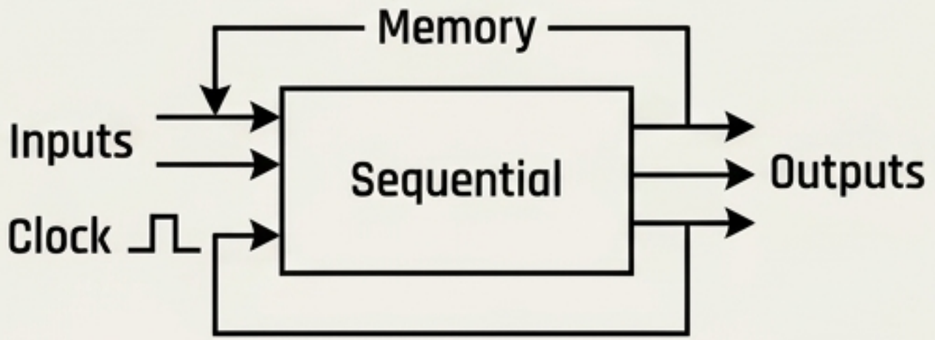
    name: XOR_Gate_Block
    inputs: [OUT_1, OUT_2]
    output: OUT_3
    config: { delay_ns: 0.8, power_mW: 1.8 }

simulation_parameters:
  test_vector_file: "tests/vectors_001.txt"
  max_cycles: 10000
  log_level: verbose

# Developer Notes:
# Ensure timing constraints are met for XOR path.
# Optimize power for low-voltage applications.
```

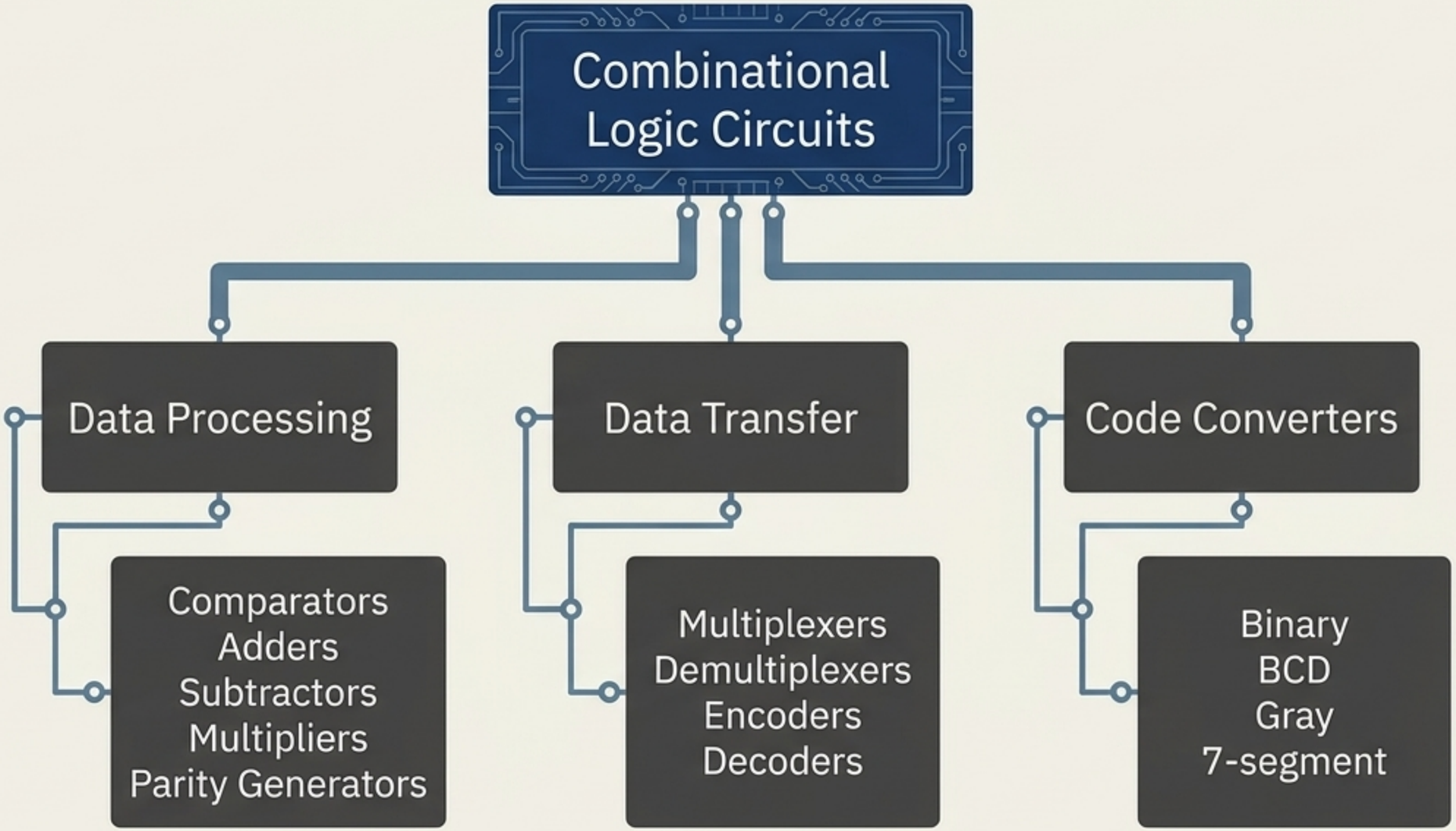
Gemini Notebook

The Logic Divide: Combinational vs. Sequential

Combinational Logic	Sequential Logic
	
Core Rule: Output depends only on present inputs.	**Core Rule:** Output depends on present inputs and past states.
State: Stateless (No memory, no feedback loop).	**State:** Stateful (Requires memory elements like flip-flops).
Timing: Governed by propagation delay; no clock.	**Timing:** Governed by clock pulses.
Examples: Adders, Subtractors, Comparators.	**Examples:** Counters, Shift Registers.

Gemini Notebook

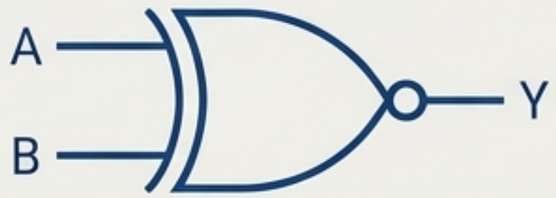
Combinational Architecture Taxonomy



Gemini Notebook

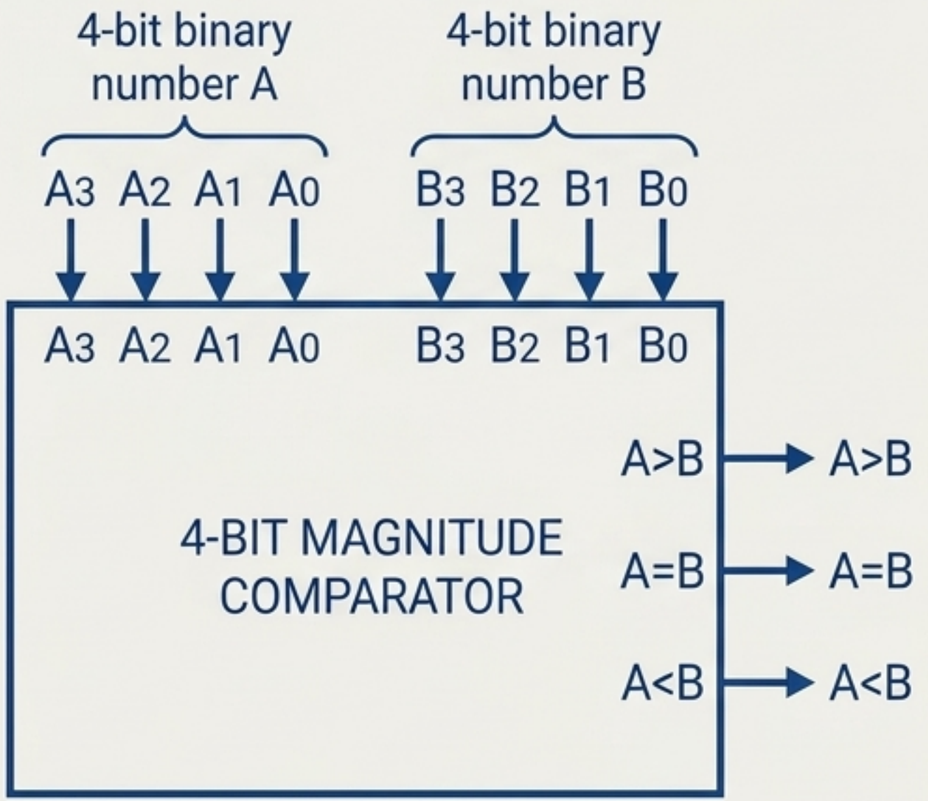
Data Processing: Binary Comparators

1-bit Equality



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

4-bit Magnitude (IC 7485)



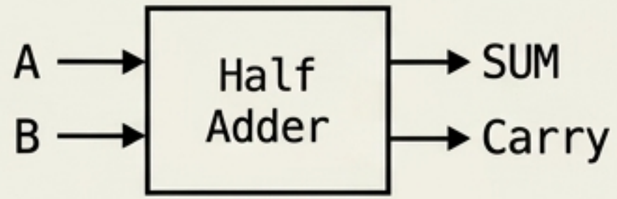
Real-World Integration

- Arithmetic Logic Units (ALU)
- Analog to Digital Converters (ADC)
- Process Control & Servo-motors
- Password & Biometric Verification

Gemini Notebook

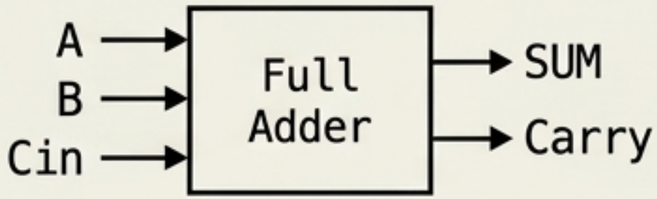
Core Arithmetic: Adders & Subtractors

Half Adder



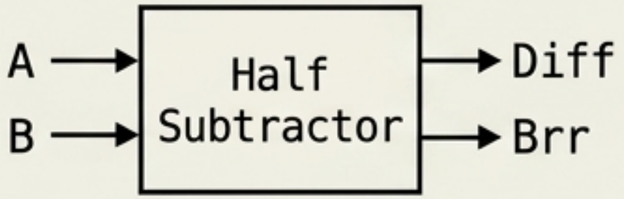
$$SUM = A \oplus B$$
$$Carry = AB$$

Full Adder



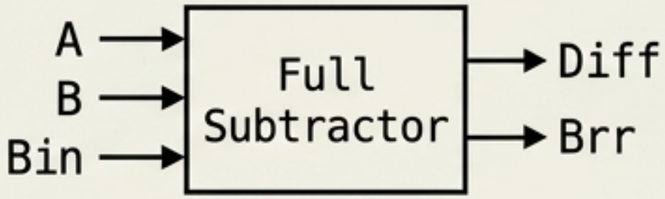
$$SUM = A \oplus B \oplus Cin$$
$$Carry = AB + BCin + ACin$$

Half Subtractor



$$Diff = A \oplus B$$
$$Brr = A'B$$

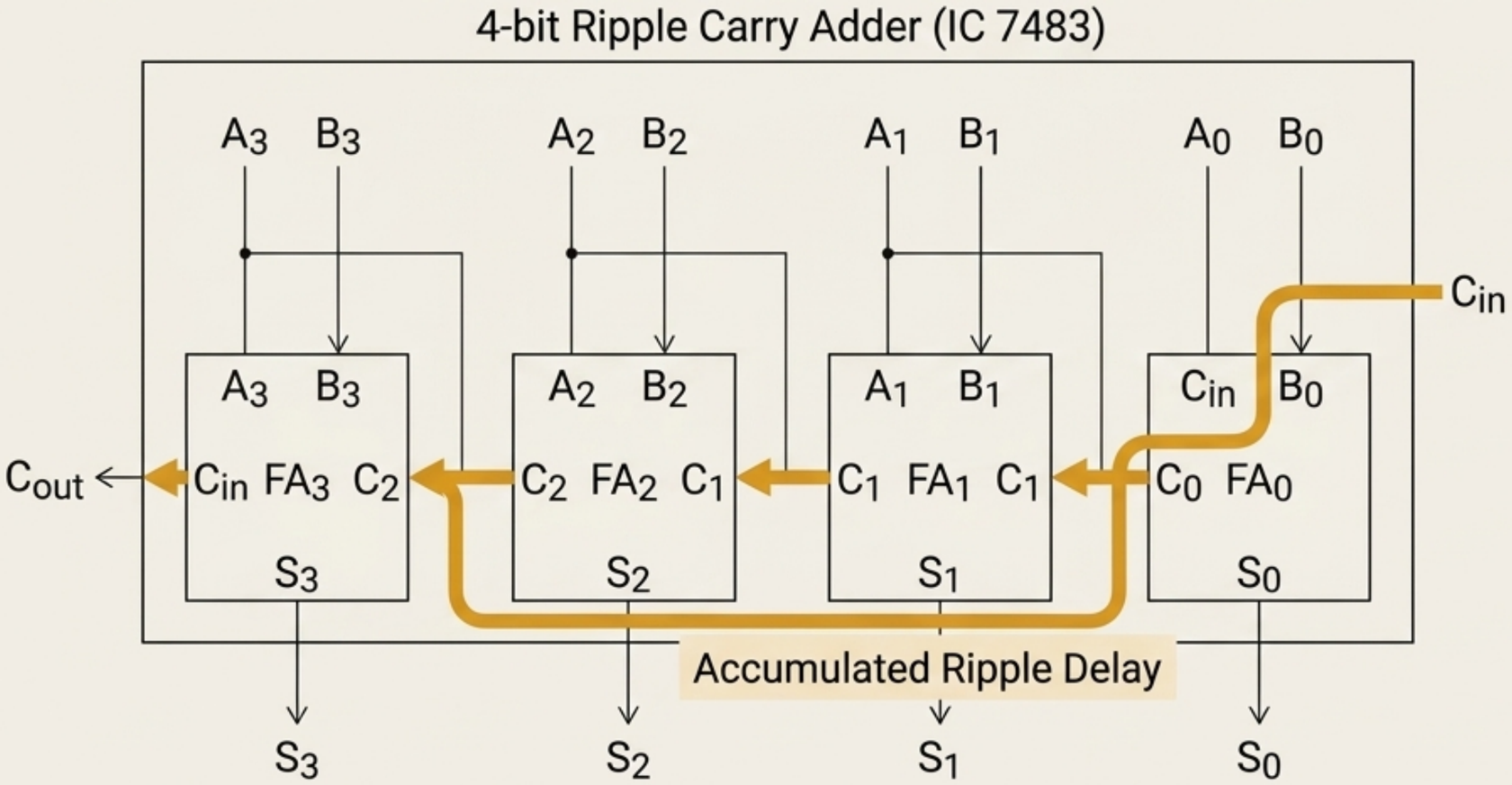
Full Subtractor



$$Diff = A \oplus B \oplus Bin$$
$$Brr = A'Bin + A'B + BBin$$

Gemini Notebook

The Scaling Limit: Propagation Delay



The MSI Solution

- Carry Lookahead Adder
- Carry Select Adder
- Carry Skip Adder
- Kogge-Stone Adder

Gemini Notebook

Enter Sequential Logic: **The Counter**

Frequency Division
Clock signal reduction & management.

Timers & Clocks
Stopwatches, system clocks, and intervals.



The Counter: A flip-flop based sequential circuit driven by a clock.

Event Counting
Digital triggers, turnstiles, and step trackers.

Memory Addressing
Sequence generation for Read/Write operations.

Gemini Notebook

Counter Taxonomy: Ripple vs. Synchronous

Asynchronous (Ripple) Counters

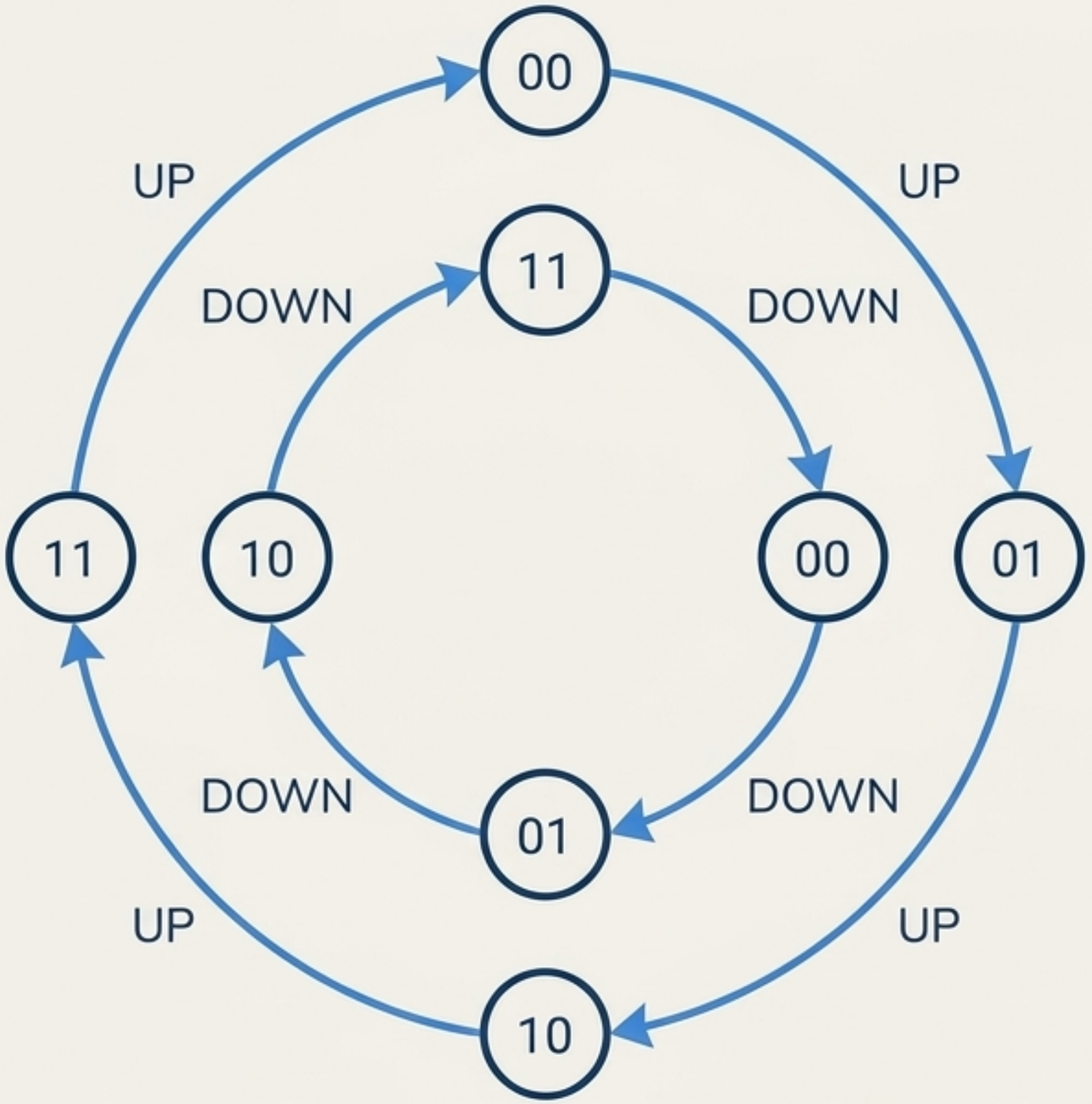
- Triggering: Sequential. The output of one flip-flop acts as the exact clock input for the next.
- Speed: Slower. Suffers from accumulated propagation delay across the circuit.
- Examples: Binary Ripple Counter, Decade Counter.

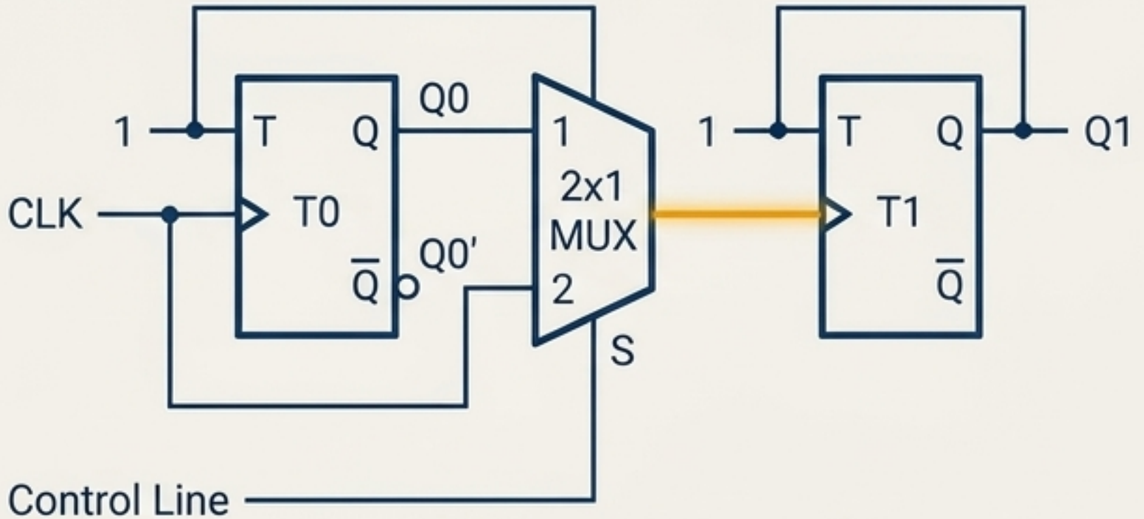
Synchronous Counters

- Triggering: Simultaneous. All flip-flops receive the exact same master clock pulse concurrently.
- Speed: Faster. Eliminates rippling delay, allowing for high-frequency operation.
- Examples: Ring Counter, Johnson (Twisted Ring) Counter.

Gemini Notebook

Directional Control: Up/Down Counters





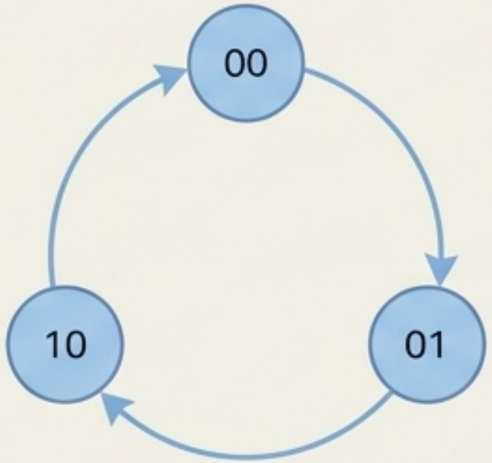

```
control_line = 0 // trigger up count sequence
control_line = 1 // trigger down count sequence
```

Gemini Notebook

Frequency Division & Mod-N Sequences

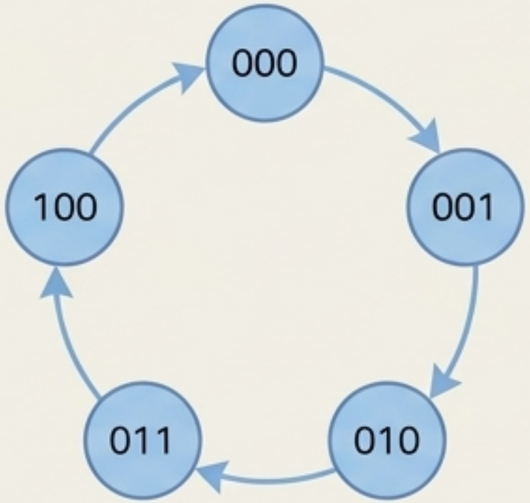
$$f_{out} = f_c / N$$ (where N is the Modulus number)

Mod-3



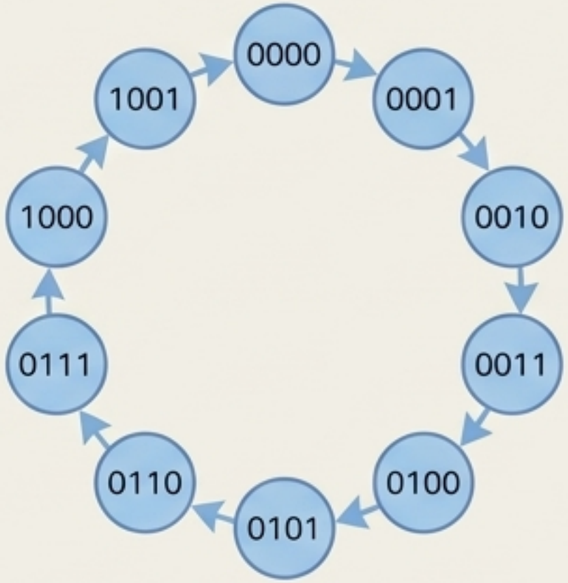
Divides frequency by 3.

Mod-5



Divides frequency by 5.

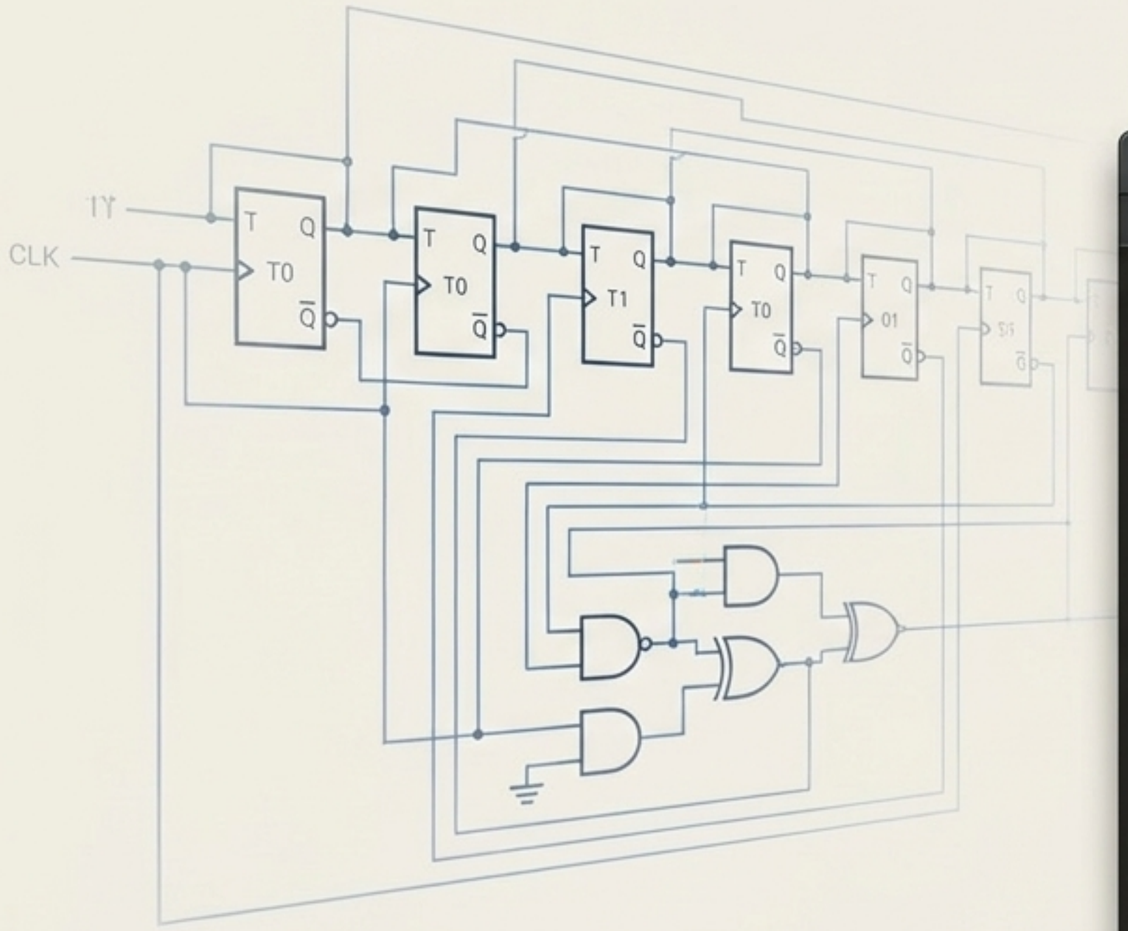
Mod-10 (Decade)



Divides frequency by 10.
Critical 50% duty cycle output.

Gemini Notebook

The Developer's Canvas: Enter GVim



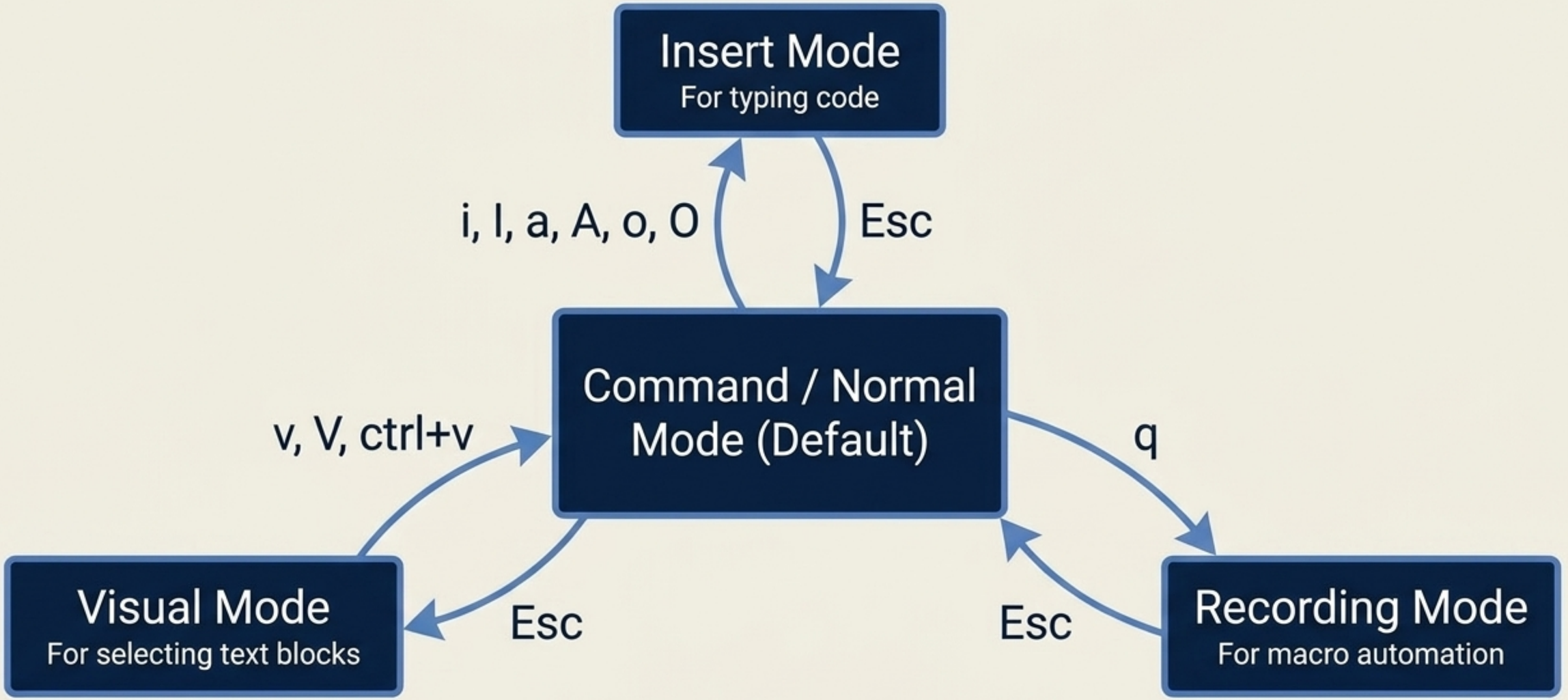
GVim is the premier graphical text editor for coding Hardware Description Languages (HDLs).

```
$ sudo apt install vim-gtk3  # Debian/Ubuntu
$ sudo yum install gvim      # RHEL/CentOS

# Essential .vimrc setup for Hardware Engineers:
set number                 # line numbers
set autoindent             # auto formatting
set tabstop=4              # 4-space tabs
syntax on                  # syntax highlighting
set guifont=consolas:h17
```

Gemini Notebook

The GVim State Machine



Gemini Notebook

Insert & Edit Mastery

State Entry (Insert)

- **i** **I** — Insert before cursor / at line start.
- **a** **A** — Append after cursor / at line end.
- **o** **O** — Open new line below / above.

Targeted Replacement

- **r** **R** — Replace single char / replace multiple.
- **s** **S** — Delete char under cursor & insert / Delete line & insert.

Deletion & Swap

- **cw** — Change word (Delete to word end, enter insert).
- **c$** — Change to end of line.
- **cc** — Change entire line.

Gemini Notebook

File Management & Buffers

I/O Operations	
Command	Action
:w :w	Write/Save file
:wq :wq	Save & Quit
:q! :q!	Force Quit without saving
:vspl <file> vspl <file>	Vertical window split
:spl <file> :spl <file>	Horizontal window split
Yanking (Copy)	
yy yy	Copy current line
nyy nyy	Copy n lines
yw yw	Copy word
Pasting & Bulk Deletion	
p / P p / P	Paste after / Paste before cursor
x x	Delete single character
dd dd	Delete current line
dG dG	Delete from cursor to end of file

Gemini Notebook